

Terraform vs Ansible Beginner Guide and Cheatsheet

Understanding when to use Terraform versus Ansible is essential for building cloud infrastructure cleanly. Terraform provisions your cloud servers from scratch, while Ansible configures software running inside them.

What you'll learn:

- 1 Spinning Up Cloud Servers with Terraform
- 2 Configuring Installed Servers with Ansible
- 3 Safely Planning Infrastructure Changes
- 4 Executing Fast Fixes with Ansible Ad-Hoc Commands

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Spinning Up Cloud Servers with Terraform

Terraform is an Infrastructure as Code tool designed to provision resources like servers and databases. You write declarative configuration files specifying what resources you want, and Terraform creates them. It tracks real infrastructure state to prevent duplicate resources. Use Terraform whenever you need to build cloud hardware.

```
$resource "aws_instance" "web_server" {  
  ami           = "ami-0c55b159cbfafa1f0"  
  instance_type = "t2.micro"  
}
```

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Configuring Installed Servers with Ansible

Ansible is a Configuration Management tool that connects to existing servers using SSH. It uses simple YAML playbooks to execute tasks like installing software packages or updating settings. Unlike Terraform, Ansible is agentless and does not require a state file. Use Ansible when your servers already exist and need software configuration.

```
$--  
- name: Install Web Server  
  hosts: webservers  
  tasks:  
    - name: Install Nginx  
      apt:  
        name: nginx  
        state: present
```

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Safely Planning Infrastructure Changes

Before applying changes to live cloud environments, Terraform lets you preview actions using plan mode. It compares your local code against actual deployed cloud resources and outputs planned additions or deletions. This safety step prevents accidental deletion of critical cloud components. Always run and review this output before deploying.

```
$ terraform plan -out=tfplan.binary
```

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Executing Fast Fixes with Ansible Ad-Hoc Commands

Writing a complete playbook takes time during operational issues. Ansible lets you run quick one-line commands directly from your terminal across hundreds of servers simultaneously. It leverages SSH keys to connect without installing software on remote servers. Use this to restart services or check system logs quickly.

```
$ansible webservers -m shell -a "systemctl status nginx"
```


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Key Takeaways

- + Use Terraform for provisioning cloud resources like VPCs, databases, and VMs.
- + Use Ansible for managing software configuration inside running servers.
- + Always execute 'terraform plan' before running 'terraform apply'.
- + Keep Ansible playbooks simple, reusable, and idempotent.
- + Combine both tools: provision infrastructure with Terraform, then configure with Ansible.

Watch Out For

- ! Using Terraform user-data scripts to handle complex software configuration.
- ! Manually configuring servers via SSH instead of tracking changes in Ansible.
- ! Skipping 'terraform plan' and accidentally destroying live cloud resources.
- ! Storing cloud access credentials inside plain text code files.

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